

Project Overview

We're building a **Cricket Player Trading Platform** where users can trade player "stocks" based on real-time cricket performance. Think of it as a stock market for cricket players, where prices fluctuate based on runs, wickets, and match performance.

Core Concept

- **Players = Stocks:** Each cricket player becomes a tradeable asset
- **Performance = Price:** Runs scored, wickets taken, catches affect stock prices
- **Real-time Trading:** Prices update live during matches
- **Profit/Loss:** Users make money based on correct predictions of player performance

Trading Types & Time Horizons

1. Day Trading (Match Day)

- **Duration:** Buy/sell within the same match
- **Settlement:** Positions close when the match ends
- **Target Users:** Active traders during live matches
- **Example:** Buy Kohli at start of innings, sell after he scores 50 runs
- **Technical Requirements:**
 - Real-time price updates every few seconds
 - Automatic position closure at match end
 - Live match integration for immediate price changes

2. Swing Trading (Test Matches Only)

- **Duration:** Hold positions for days to weeks
- **Use Case:** Multi-day Test matches
- **Target Users:** Strategic long-term traders
- **Example:** Buy spinner's stock before Day 4-5 when pitch deteriorates
- **Technical Requirements:**
 - Position tracking across multiple days
 - Overnight price adjustments
 - Test match-specific scoring algorithms

3. Position Trading (Tournament/Series)

- **Duration:** Hold for weeks to months
- **Use Case:** IPL seasons, World Cup, bilateral series

- **Target Users:** Season-long investors
- **Example:** Buy emerging player's stock at IPL start, hold entire tournament
- **Technical Requirements:**
 - Long-term position management
 - Tournament-specific pricing models
 - Performance tracking across multiple matches

4. Scalping (Ultra Short-term)

- **Duration:** Seconds to minutes
- **Use Case:** Per over or per ball trading
- **Target Users:** High-frequency traders
- **Example:** Buy bowler's stock before power-play, sell after tight over
- **Technical Requirements:**
 - Sub-second price updates
 - Ball-by-ball data integration
 - High-frequency order matching system

5. Long Trading (Buy First)

- **Mechanism:** Purchase player stock expecting price increase
- **Profit:** When player performs well, sell at higher price
- **Risk:** Limited to investment amount
- **Example:** Buy batsman's stock expecting big innings

6. Short Trading (Sell First)

- **Mechanism:** Sell player stock expecting price decrease
- **Profit:** When player performs poorly, buy back at lower price
- **Risk:** Potentially unlimited if player performs exceptionally
- **Example:** Short a bowler's stock expecting expensive bowling figures

Core Systems Needed:

1. **Live Match Data Integration**
 - Ball-by-ball updates
 - Score tracking
 - Player statistics
2. **Dynamic Pricing Engine**
 - Real-time price calculations
 - Different algorithms for each trading type

- Risk management controls
- 3. **Order Management System**
 - Buy/sell order matching
 - Position tracking
 - Portfolio management
- 4. **Risk Management**
 - Position limits
 - Margin requirements for short trading
 - Circuit breakers for extreme volatility